

BIOL-1 Exam 02 Answer Key

Part A: Multiple Choice (30 points)

1 point each.

Module 07: Molecular Genetics

1. **C** (DNA → RNA → Protein)
2. **C** (Adenine pairs with thymine)
3. **B** (A in DNA pairs with U in RNA during transcription)
4. **C** (DNA polymerase)
5. **B** (Ribosomes in the cytoplasm / on rough ER)
6. **B** (Frameshift mutation)

Module 08: Cellular Genetics

1. **B** (Haploid, $n = 23$)
2. **B** (Two genetically identical diploid daughter cells)
3. **B** (Homologous chromosomes separate in meiosis I)
4. **B** (Prophase I of meiosis)
5. **B** (Random alignment creates many allele combinations)
6. **B** (Failure of proper separation; may yield aneuploidy)

Module 09: Inheritance Genetics

1. **B** (Alleles are alternative versions of the same gene)
2. **A** (Bb is genotype; purple flowers are phenotype)
3. **C** (3 : 1 phenotypic ratio for complete dominance)
4. **B** (1 : 2 : 1 with a blended heterozygote in incomplete dominance)
5. **B** (Codominance for type AB)
6. **B** (Two unaffected parents can have an affected child—classic autosomal recessive clue)

Module 10: Epigenetics

1. **B** (Gene expression changes without DNA sequence change)
2. **B** (Silences / turns gene off or down)
3. **B** (Looser chromatin; favors gene activity / turning genes on)
4. **B** (Mutation changes sequence; epigenetics changes use of same sequence)
5. **B** (Barr body)
6. **B** (Different cell types express different genes)

Module 11: Genomics & Biotechnology

1. **B** (Amplify / copy a targeted DNA region)
 2. **B** (Toward positive electrode; smaller fragments usually travel farther)
 3. **B** (Universal / nearly universal genetic code allows translation of the gene)
 4. **B** (STR alleles inherited from parents)
 5. **B** (Targeted DNA cut / gene editing)
 6. **B** (DNA ligase joins fragments)
-

Part B: Fill in the Blank (11 points)

1 point each. Minor spelling errors are acceptable as long as the intent is clear.

1. **Transcription**
 2. **Translation**
 3. **Anticodon**
 4. **Methylation** (accept DNA methylation)
 5. **Epigenetics**
 6. **Heterozygous**
 7. **Nondisjunction**
 8. **Codominance**
 9. **Plasmid**
 10. **Restriction** (accept restriction enzyme)
 11. **Diploid**
-

Part C: Free Response (9 points)

Choose *THREE* out of five. Each question is worth 3 points.

Award partial credit as appropriate for incomplete but partially correct concepts.

1. Information Flow and Regulation.

- *Key concepts:* **Transcription** (DNA → mRNA, nucleus in eukaryotes); **Translation** (mRNA → protein on ribosomes); mention tRNA/amino acids if present.
- *Regulation hook:* **Epigenetics** (methylation, histones) or **gene regulation** / **transcription factors** turning genes on or off **without changing DNA sequence**; different cell types use different subsets of genes.

2. Mitosis vs. Meiosis.

- **Mitosis:** one division → **2** daughter cells; **diploid** (2n) in humans for somatic cells; genetically **identical** (barring mutation); purposes include **growth and repair**.
- **Meiosis:** two divisions → **4** cells; **haploid** (n) gametes; **genetically different** from each other (crossing over, independent assortment, random fertilization); purpose is **gamete production** / **sexual reproduction**.

3. Inheritance in Words.

- **Genotype:** allele combination (e.g., RR, Rr, rr). **Phenotype:** observable trait.
- **Incomplete dominance:** heterozygote is **intermediate** / **blended** (e.g., red × white → pink), not a clear dominant masking recessive in the same way as complete dominance.

4. Epigenetics vs. Mutation.

- **Epigenetics:** changes **gene expression** (on/off, more/less) **without** changing the DNA letter sequence; often reversible in principle.
- **Mutation:** change in **DNA sequence**.

- *Environment example:* **diet** (methyl donors), **chronic stress**, **exercise**, or **toxins** altering regulation—any one explained briefly is sufficient.

5. Biotechnology Toolkit.

- **PCR path:** rapidly **amplifies** a specific DNA segment (many copies from a tiny sample); connects to **denature** / **prime** / **extend** logic at intro level.
- **Gel path:** separates fragments **by size**; **smaller** fragments migrate farther; compares **band patterns** (e.g., STR profiles).
- **Plasmid / universal code path:** foreign gene inserted into **plasmid**; host reads gene because **codons specify the same amino acids** across species (universal genetic code), so **transcription/translation** can produce the protein.

End of exam key.